

Light-Curve Fit Report

LCURVE — Levenberg–Marquardt Solver

config.json

April 26, 2026

1. Fit Summary

Converged — xtol (parameter change small)

Iterations	86	χ^2_{LC}	0.0001
Function evaluations	417	χ^2_{red}	0.0000
N_{data}	50	$\ r\ ^2$	0.0001
N_{par}	6	N_{prior}	8
P_{orb} (d)	0.69292600		

2. Best-Fit Parameters

Parameter	Value $\pm \sigma$	Limits
q	0.976 ± 0.033	[0.0100, 100.0000]
i (deg)	77.7 ± 4.3	[0.0000, 90.0000]
r_1	0.03984 ± 0.00065	[0.0000, <i>inf</i>]
r_2	0.0044 ± 0.0018	[0.0000, <i>inf</i>]
v_{scale} (km/s)	260.5 ± 2.7	[0.0000, <i>inf</i>]
T_0	0.759 ± 0.017	[<i>-inf</i> , <i>inf</i>]

3. Derived Physical Quantities

Quantity	Implied $\pm\sigma$	Observed	Pull
K_1 (km/s)	125.7 ± 1.3	$125.0^{+3.5}_{-3.5}$	0.20σ
K_2 (km/s)	128.8 ± 3.3	—	—
R_1 ($R_\odot$)	0.1422 ± 0.0019	$0.1420^{+0.0050}_{-0.0050}$	0.03σ
R_2 ($R_\odot$)	0.0158 ± 0.0064	—	—
M_1 ($M_\odot$)	0.642 ± 0.018	$0.630^{+0.050}_{-0.050}$	0.25σ
M_2 ($M_\odot$)	0.627 ± 0.025	—	—
M_{total} ($M_\odot$)	1.269 ± 0.039	—	—
q	0.976 ± 0.033	—	—
$\log g_1$ (dex)	5.940 ± 0.016	$5.94^{+0.10}_{-0.10}$	0.00σ
$\log g_2$ (dex)	7.84 ± 0.35	$7.5^{+1.0}_{-1.0}$	0.34σ
T_1 (K)	30492.0000	30492^{+1000}_{-1000}	0.00σ
T_2 (K)	5000.0000	5000^{+2000}_{-2000}	0.00σ
a ($R_\odot$)	3.568 ± 0.036	—	—

4. Prior Constraint Residuals

Residuals include prior weighting and balance scaling.

Constraint	Scaled Residual	Status
K1	0.001σ	OK
M1	0.001σ	OK
M2min	-0.001σ	OK
R1	0.000σ	OK
T1	0.000σ	OK
T2	0.000σ	OK
logg1	0.000σ	OK
logg2	0.001σ	OK

5. Correlation Matrix

	q	i	r_1	r_2	v_s	T_0
q	1	-0.76	-0.26	0.08	0.40	0.04
i	-0.76	1	0.47	-0.09	-0.81	-0.05
r_1	-0.26	0.47	1	-0.04	-0.56	-0.02
r_2	0.08	-0.09	-0.04	1	0.07	0.00
v_s	0.40	-0.81	-0.56	0.07	1	0.04
T_0	0.04	-0.05	-0.02	0.00	0.04	1